

to do and nobody undoes the work of another.

6406531500236. ✖ No matter what industry you work in, build a culture of trust, where people cheer and motivate one another in working towards one common goal.

Question Number : 36 Question Id : 640653450756 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

You are the chairman of your colony association, which consist of volunteer members who are residents in your neighbourhood. At a recent meeting, you delegated a substantial amount of work to some of the other members. "I know these are difficult tasks, but you're part of this team and you need to get this work done", you told them. But at your next monthly meeting, you were surprised those association members weren't there as they simply decided to quit because they had better things to do with their time. Which aspect of teamwork is missing in this example?

Options :

- 6406531500253. ✔ Adequate task definition
- 6406531500254. ✖ Managerial interference
- 6406531500255. ✔ Collaboration
- 6406531500256. ✖ Lack of leadership

AI

Section Id :	64065329293
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	25

Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065364736
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 37 Question Id : 640653450762 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: AI: SEARCH METHODS FOR PROBLEM SOLVING"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECT TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT ,PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

- Options :
- 6406531500277.  Yes
 - 6406531500278.  No

Sub-Section Number :	2
Sub-Section Id :	64065364737
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450763 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Question Numbers : (38 to 39)

Question Label : Comprehension

SEARCH

Consider a 5-4-1 water jug puzzle with three jugs a, b, c of capacity 5L, 4L, 1L, respectively. A state is represented as a three digit string ABC, where A, B, C denote the volume of water present in the jugs a, b, c, respectively. For example, 140 denotes a state where jug a has 1 liter of water, jug b has 4 liters of water and jug c is empty.

Two types of moves (state transitions) are allowed:

xEy: pick up jug x and empty it into jug y,

xFy: pick up jug x and fill up (top up) jug y.

So the possible moves are: aEb, aEc, aFb, aFc, bEa, bEc, bFa, bFc, cEa, cEb, cFa, cFb.

Tie-breaker: in some states, both xEy and xFy may produce the same output state, in such cases, choose xEy to break the tie. For example, from the 140 state, both aEc and aFc produce the same output state 041, therefore, to break the tie choose aEc as the move.

For the 5-4-1 water jug puzzle, the MoveGen function takes a state ABC and returns a set of states that can be produced in one move. For example, MoveGen(500) returns {140, 401 }.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 38 Question Id : 640653450764 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

Let 500 be the start state and 221 be the goal state, find the shortest path from start to goal. Enter the path starting with 500 as a comma separated list of states.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 500,411,747

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

500,140,131,230,221

Question Number : 39 **Question Id :** 640653450765 **Question Type :** SA **Calculator :** None

Response Time : N.A **Think Time :** N.A **Minimum Instruction Time :** 0

Correct Marks : 1

Question Label : Short Answer Question

Let 500 be the start state and 221 be the goal state, find the sequence of moves that produce the shortest path from start to goal. Enter the sequence of moves as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: aFc,cFb,aEc

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

aFb,bFc,cEa,bFc

Sub-Section Number : 3

Sub-Section Id : 64065364738

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 40 Question Id : 640653450766 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

SEARCH FOR OPTIMAL SOLUTION

Under what cases does Depth First Iterative Deepening (DFID) guarantee an optimal solution?

Options :

- 6406531500281. ✖ When it inspects only new nodes
- 6406531500282. ✖ When it inspects new as well as open nodes
- 6406531500283. ✔ When it inspects new as well as closed nodes
- 6406531500284. ✖ None of these

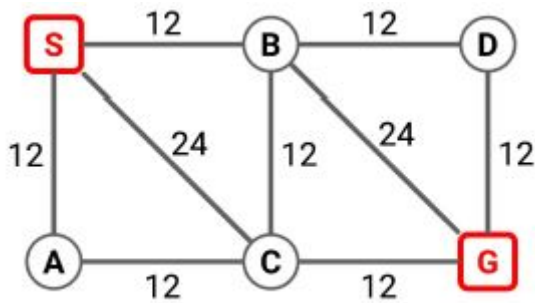
Question Number : 41 Question Id : 640653450770 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

SEARCH FOR OPTIMAL SOLUTION

In the map, nodes are placed on a grid where each tile is 10x10 in size, is the Manhattan distance heuristic admissible?



Options :

6406531500305. ☒ Yes

6406531500306. ☐ No

6406531500307. ☐ Cannot be determined

Sub-Section Number :

4

Sub-Section Id :

64065364739

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 42 Question Id : 640653450767 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

SEARCH FOR OPTIMAL SOLUTION

Given a finite state space with unit edge costs, and a heuristic function whose properties are not known, which of the following algorithms are suitable for finding the optimal path?

Options :

6406531500285. ☐ Depth First Search

6406531500286. ☒ Breadth First Search

6406531500287. ☒ Dijkstra's algorithm

6406531500288. ☒ Branch and Bound Search

6406531500289. ☐ A*

6406531500290. ☐ WA* for some w

6406531500291. ☐ Sparse Memory Graph Search (SMGS)

Question Number : 43 Question Id : 640653450768 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

SEARCH FOR OPTIMAL SOLUTION

Given a state space with Euclidean edge costs, and with the best admissible heuristics, which of the following algorithms will find the optimal path?

Options :

- 6406531500292. ✖ Depth First Search
- 6406531500293. ✖ Breadth First Search
- 6406531500294. ✔ Dijkstra's algorithm
- 6406531500295. ✔ Branch and Bound Search
- 6406531500296. ✔ A*
- 6406531500297. ✖ WA* for some w
- 6406531500298. ✔ Sparse Memory Graph Search (SMGS)

Question Number : 44 Question Id : 640653450769 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

SEARCH FOR OPTIMAL SOLUTION

Which of the following are true about the Branch-and-Bound method (discussed in the lecture) that is used for finding optimal tours for the Traveling Salesman Problem?

Options :

- 6406531500299. ✔ The root of the search tree represents all possible tours.
- 6406531500300. ✖ The root of the search tree represents the shortest/optimal tour.

6406531500301. ✔ Each node in the search tree represents a set of tours.

6406531500302. ✖ Each node in the search tree represents a single tour.

6406531500303. ✔ The costs of the search tree nodes monotonically INCREASE from the root to leaf.

6406531500304. ✖ The costs of the search tree nodes monotonically DECREASE from the root to leaf.

Sub-Section Number :	5
Sub-Section Id :	64065364740
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450771 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

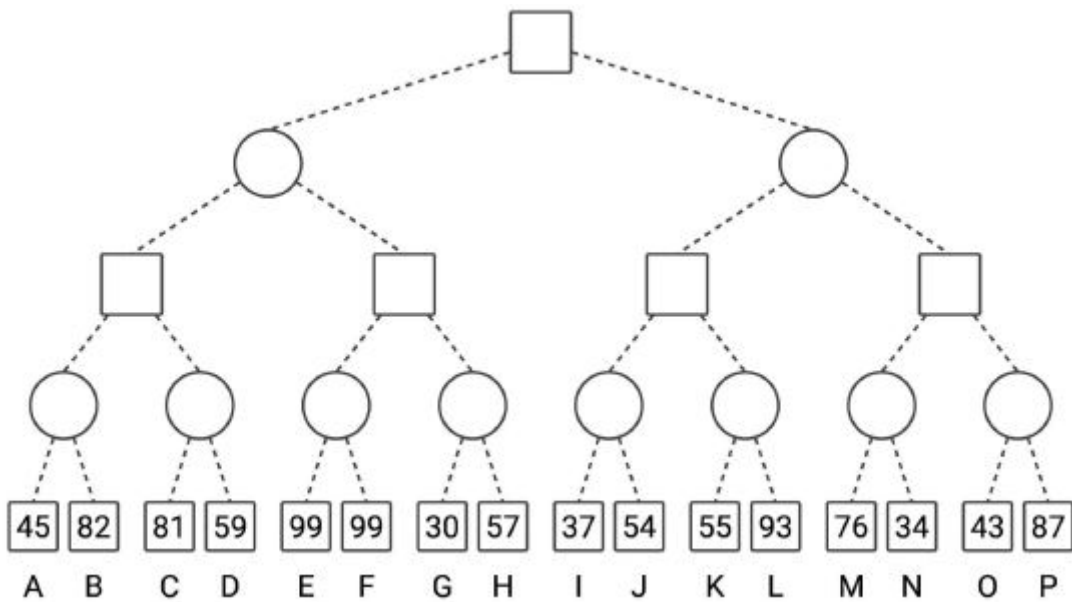
Question Numbers : (45 to 47)

Question Label : Comprehension

GAMES

The figure shows a 4-ply game tree with evaluation function values at the horizon. The nodes in the horizon are assigned labels A,B,C,...,P. Use these labels when asked to enter a horizon node or a list of horizon nodes.

Tie-breaker: when several nodes qualify then select the left most node, if tie persists then select the deepest node among the left most nodes.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 45 Question Id : 640653450772 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following are strategies for MAX?

Options :

6406531500308. ✓ A,B,E,F

6406531500309. ✓ C,D,G,H

6406531500310. ✗ A,C,I,J

6406531500311. ✗ B,D,J,L

Question Number : 46 Question Id : 640653450773 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the best strategy for MAX. Enter the nodes in the ascending order of node labels.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

C,D,E,F

Question Number : 47 **Question Id** : 640653450774 **Question Type** : SA **Calculator** : None

Response Time : N.A **Think Time** : N.A **Minimum Instruction Time** : 0

Correct Marks : 1

Question Label : Short Answer Question

List the first five horizon nodes pruned by Alpha-Beta.

Enter a comma separated sorted list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

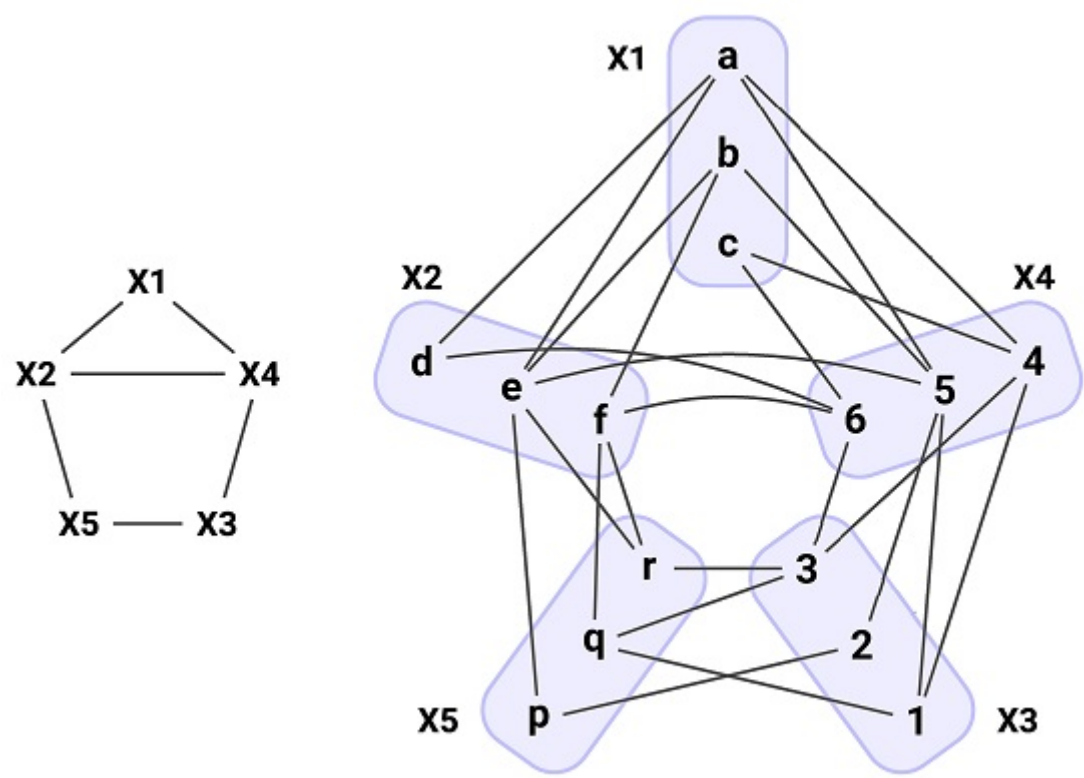
G,H,J,L,M

Question Id : 640653450789 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Question Numbers : (48 to 49)

Question Label : Comprehension

CONSTRAINT SATISFACTION

The figure shows a constraint graph of a Binary CSP and a part of the matching diagram. When a pair of variables (like, X1 and X5) do not have a constraint in the constraint graph then assume a UNIVERSAL RELATION in the matching diagram.



Process the variables, and their values, in ALPHABETIC order

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 48 Question Id : 640653450790 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

The Forward-Checking algorithm begins by assigning $X_1=a$. What are the NEXT four values assigned to variables by the Forward-Checking algorithm? List the values as a comma separated list in the order they are assigned.

Enter a comma separated list of values.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: a,6,1,p

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

d,e,1,2

e,2,5,p

Question Number : 49 Question Id : 640653450791 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the first solution found by the Forward-Checking algorithm?

Enter a comma separated list of values.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: a,6,1,p

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

a,e,2,5,p

Sub-Section Number :	6
Sub-Section Id :	64065364741
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450775 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (50 to 53)

Question Label : Comprehension

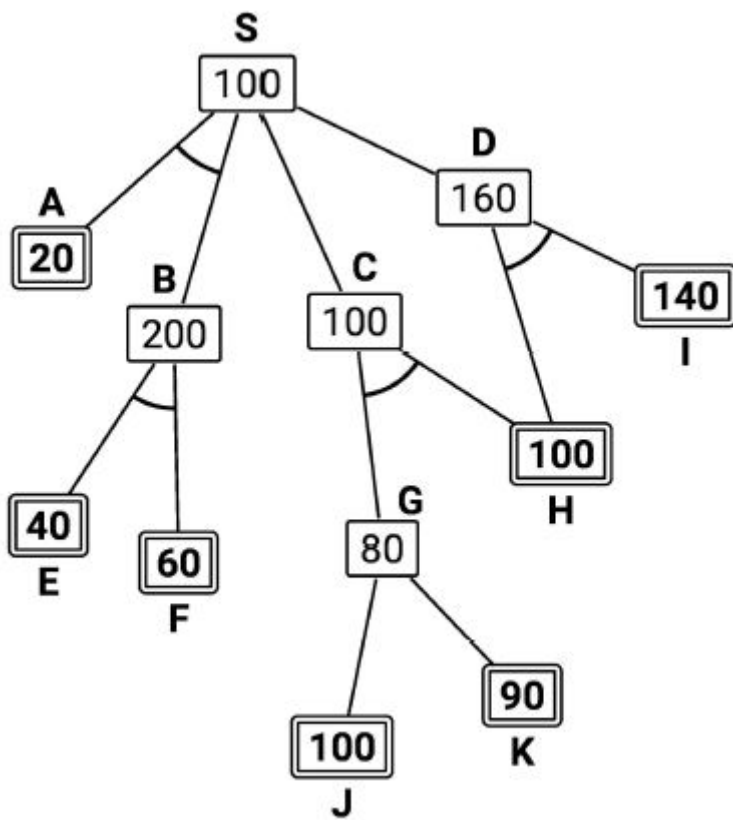
PROBLEM DECOMPOSITION

The figure shows an AND-OR graph that depicts how a problem S can be decomposed and transformed into one or more simpler problems. Nodes are uniquely identified by labels (S, A, B, ...). The number in each node is the heuristic estimate of the cost of solving that node.

Nodes with double lines are primitive nodes and their values are actual costs. Observe that a primitive node is added to the graph by its parent when the parent is expanded, and the primitive node is labeled as SOLVED and it will not be expanded subsequently.

Tie-breaker 1: For nodes with the same cost, expand in the ascending order of node labels.

Tie-breaker 2: For AND nodes, expand the unsolved branch with the highest cost.



Each edge has a unit cost. Solve for S, then answer the given subquestions

Sub questions

Question Number : 50 Question Id : 640653450776 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

List the first three nodes (including S) expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,C,D

S,C,D,G

Question Number : 51 Question Id : 640653450777 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Determine the value of the start node S after each node expansion. List the first three values of S that correspond to the first three node expansions.

Enter a comma separated list of numbers.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

101,161,183

101,161,183,194

Question Number : 52 Question Id : 640653450778 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the final value of the start node S?

Enter a number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

194

Question Number : 53 Question Id : 640653450779 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

In the given AND-OR graph, is the heuristic admissible?

Options :

6406531500317. ✓ No

6406531500318. ✗ Yes

6406531500319. ✗ Cannot be determined

Question Id : 640653450780 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

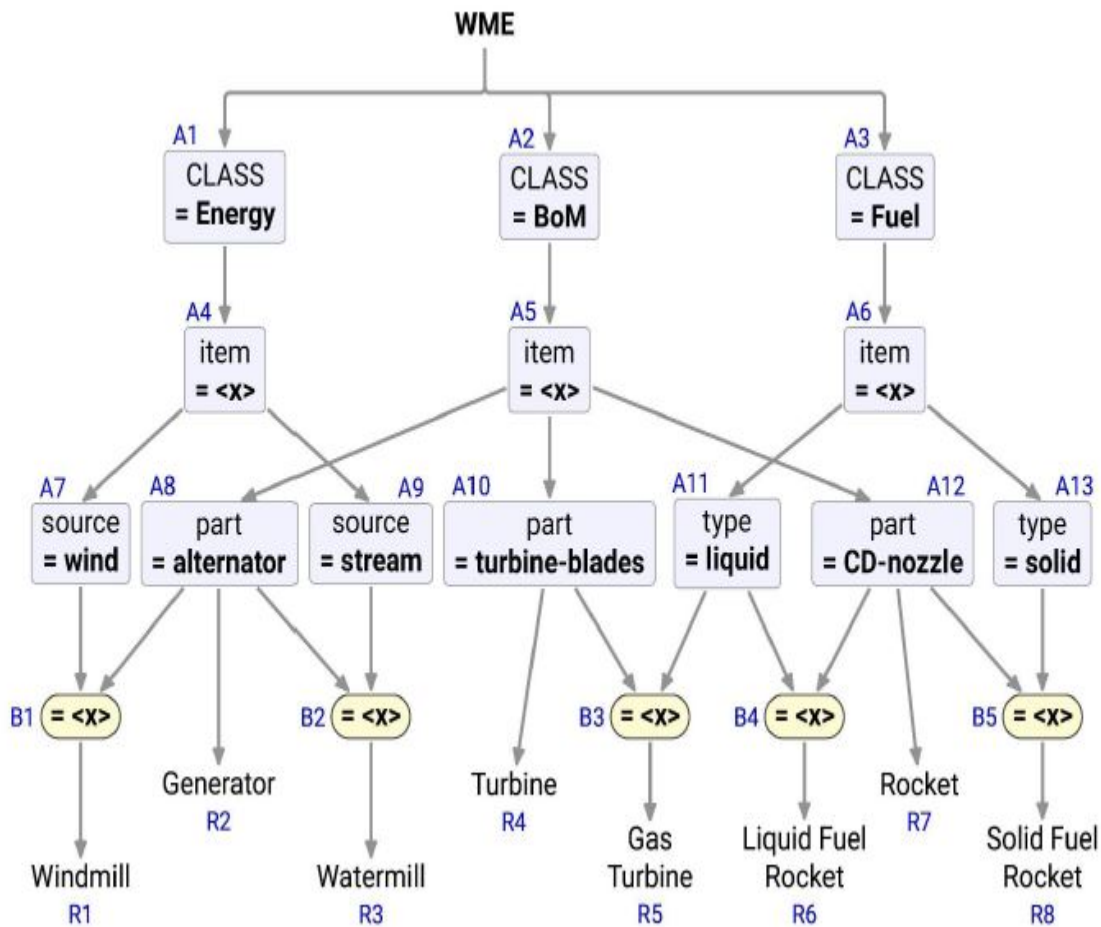
Question Numbers : (54 to 56)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A Rete Net for classification of machines is shown in the figure. The labels A1, A2, A3, ..., A10, A11, A12, A13, ..., B1, B2, ..., B5, R1, R2, ..., R8, uniquely identify the nodes in the network. When

required, use the above label ordering to **break ties** and to enter your answers.



Run the Rete algorithm for the Working Memory shown below, where the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

101. (Fuel ^item machine5 ^type solid)
102. (Fuel ^item machine2 ^type liquid)
103. (Fuel ^item machine3)
104. (BoM ^item machine4 ^part alternator)
105. (Energy ^item machine4 ^source wind)
106. (BoM ^item machine3 ^part alternator)
107. (BoM ^item machine1 ^part CD-nozzle)

For each WME identify its location (node label) in the Rete Net, and prepare the conflict set for the first cycle, then answer the given subquestions.

Sub questions

Question Number : 54 Question Id : 640653450781 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples are in the conflict-set?

Options :

6406531500320. ✓ R1,104,105

6406531500321. ✓ R2,104

6406531500322. ✓ R2,106

6406531500323. ✓ R7,107

6406531500324. ✗ R1,104,106

6406531500325. ✗ R3,104,106

6406531500326. ✗ R6,102,107

Question Number : 55 Question Id : 640653450782 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

If the Inference Engine uses **Specificity** as the conflict resolution strategy then identify the rule-data tuples that will be ready to fire. If multiple rule-data tuples qualify then choose one.

Enter a comma separated list with a rule label followed by timestamps in sorted order.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: R1,101,102,103

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

R1,104,105

R1,105,104

Question Number : 56 Question Id : 640653450783 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

If the Inference Engine uses **Recency** as the conflict resolution strategy then identify the rule-data tuples that will be ready to fire. If multiple rule-data tuples qualify then choose one.

Enter a comma separated list with a rule label followed by timestamps in sorted order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: R1,101,102,103

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

R7,107

Question Id : 640653450784 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (57 to 60)

Question Label : Comprehension

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X). Pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X). Place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y). Pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

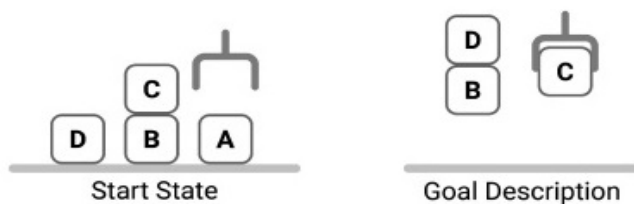
Stack(X,Y). Place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

Consider the following planning problem.



Start: { onTable(A), onTable(B), onTable(D), on(C,B), clear(A), clear(C), clear(D), armEmpty }

Goal: { on(D,B), holding(C) }

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 57 Question Id : 640653450785 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for the given problem?

Options :

6406531500329. ✓ Pickup(A)

6406531500330. ✓ Pickup(D)

6406531500331. ✓ Unstack(C,B)

6406531500332. ✗ Pickup(C)

6406531500333. ✗ Stack(D,B)

6406531500334. ✗ Unstack(D,B)

Question Number : 58 Question Id : 640653450786 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions for the given problem?

Options :

6406531500335. ✗ Pickup(A)

6406531500336. ✗ Pickup(D)

6406531500337. ✓ Unstack(C,B)

6406531500338. ✓ Pickup(C)

6406531500339. ✓ Stack(D,B)

6406531500340. ✗ Unstack(D,B)

Question Number : 59 Question Id : 640653450787 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0